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Rajarata University of Sri Lanka, Mihintale
B.Sc. in Applied Sciences

STA 1202 – Introduction to Probability Theory
Tutorial No: 2

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1. A fair coin is tossed three times and a note is taken of the number of tails.

a. List the possible outcomes.
b. List the possible values of the random variable X, representing the number of tails obtained in the three tosses.
c. Find the probability distribution of X.
d. Find $Pr(X \leq 2)$.

2. Find the value of k for each of the following discrete probability distributions.

a.

x	1	2	3	4	5
Pr(X=x)	0.3	0.2	0.2	k	0.1

b.

x	1	2	3	4	5
Pr(X=x)	0.1	0.1	0.1	0.1	k

c.

x	1	2	3	4	5
Pr(X=x)	k	2k	3k	4k	k

3. Three balls are selected from a box containing 4 red balls and 5 blue balls. If the ball chosen after each selection is replaced before the next selection, find, correct to 4 decimal places:

a. the probability distribution for the number of red balls drawn:
   i. 0 red balls
   ii. 1 red ball
   iii. 2 red balls
   iv. 3 red balls

b. the probability that three reds are chosen, given that at least one ball is red.

4. The probability function of the discrete random variable X is shown in the table. Given that $E(X) = 2.95$, find the values of "a" and "b".

x	1	2	3	4
Pr(X=x)	0.2	a	0.25	b

5. Christian decides to apply for a job selling mobile phones. He receives a base salary of \$180 per month and \$12 for every mobile phone sold. The following table shows the probability of a particular number of mobile phones, x, being sold per month. What would be the expected salary Christian would receive each month?

x	50	100	150	200	250
Pr(X=x)	0.32	0.38	0.2	0.06	0.04

6. In order to encourage carpooling, a new toll is to be introduced on the International Gateway. If the car has no passengers, a toll of \$2 applies. Cars with one passenger pay a \$1.50 toll, cars with two passengers pay a \$1 toll and cars with 3 or more passengers pay no toll. Long-term statistics show that the number of passengers follows the probability distribution given below.

x	0	1	2	≥ 3
Pr(X=x)	0.5	0.3	0.15	0.05

a. Construct a probability distribution of the toll paid.
b. Find the mean toll paid per car.
c. Find the standard deviation of tolls paid.

7. A random variable has the following probability distribution.

x	0	1	3	5	7
Pr(X=x)	0.27	0.15	0.13	0.1	0.35

Find:

a. $Var\ (X)$
b. $Var\ (3X)$
c. $Var\ (10X - 5)$
d. $Var\ (5X - 2)$

8. If X is a continuous random variable with the probability density function defined as

$$f(X) = \begin{cases} \frac{1}{2}x - 4, & 8 \leq x \leq 10 \\ 0 \quad , & elsewhere \end{cases}$$

Find

a. Pr $(X > 9)$
b. Pr $(8.5 < X < 9)$
c. Pr $(X > 9 \mid X > 8.5)$

9. The probability density function of the age of the babies, x years, being brought to a post antenatal clinic is given by

$$f(X) = \begin{cases} \frac{3}{4}x(2-x), & 0 < x < 2 \\ 0 \quad , & elsewhere \end{cases}$$

i. If 60 babies are brought in on a particular day, how many are expected to be under 8 months old?
ii. Find the mean age of the babies brought to the clinic?
iii. For the baby's distribution, find the variance of x.

10. i. find the mode of the p.d.f defined by

$$f(x) = 12x^2(1-x), \; 0 < x < 1$$

ii. find the median of this p.d.f

$$f(x) = \frac{x}{4}, \quad 1 < x < 3$$

(Please answer question number 1, 2) a, b , 3, 5, 6, 7, 9 and 10 and submit the answer script to the tutorial room on or before 27th of November 2023)